# Computational Communication Science 2
# Course Manual

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# Chapter 1

# About this course

This course manual contains general information, guidelines, rules and schedules for the course *Computational Communication Science 2* (6 ECTS), part of the *Communication in the Digital Society Minor* offered by the College of Communication at the University of Amsterdam. Please read it carefully, as it contains important information regarding assignments, deadlines and grading.

## 1.1 Course description

A sales website that recommends new products to you personally, a company that uses a chatbot to answer your questions, or an algorithm that automatically identifies and warns you about fake news content: in our digital society, computational methods increasingly shape how we communicate and interact with information.

In this course, we examine the computational methods that underlie these forms of communication. We explore the basic principles behind their design, acquire an understanding of their implications, and learn how these methods can be used for scientific research.

Lectures discuss computational methods and their societal implications. Tutorials focus on hands-on exercises in Python.

At the end of the course, you will have a basic understanding of computational methods used in communication research and practical experience applying them.

This course consists of one lecture and one tutorial per week. In total there are 28 contact hours ($7 \times$ 2-hour lectures and $7 \times$ 2-hour tutorials).

## 1.2 Goals

Upon completion of this course, students should:

a) Understand key computational techniques used to study communication in the digital society.

b) Understand how rule-based, unsupervised, and supervised techniques can be applied to communication research.

c) Identify benefits and limitations of different computational methods.

d) Understand which communication science questions can be addressed with computational methods.

e) Apply a subset of these techniques independently.

f) Gain experience solving problems in Python using online resources.

g) Clearly communicate methodological steps taken in computational research.

## 1.3 Study materials

Python is the programming language used in this course. Students should bring a laptop to class with a working Python environment installed.

Assigned readings are listed in this syllabus and are available through the UvA Digital Library or Google Scholar. If a reading is not available online, it will be provided via Canvas.

# Chapter 2

# Assignments, grading, and rules

Assessment is based on a combination of individual and group assignments.

## 2.1 Overview of assessments

- Regular multiple-choice questions (20%)
- Group assignment: Written report (20%)
- Group assignment: Presentation (15%)
- Exam (45%)

## 2.2 Regular multiple-choice questions (20%)

At the start of several lab sessions, students will answer four questions about the assigned literature and the preceding lecture.

MC questions are administered in weeks 1, 3, 4, 7, and 8. In total students answer 20 questions.

To receive full marks, at least 16 questions must be answered correctly.

## 2.3 Group assignment: report (20%)

Students work in groups on a computational research project. See Section 3 for details.

## 2.4 Group assignment: presentation (15%)

In week 4 (Tuesday 21–4), each group presents their project and explains their methodology, code, and findings.

## 2.5 Exam (45%)

During the final tutorial session, students complete an exam testing conceptual understanding and including a coding task.

## 2.6 Grading

Students must obtain a pass (5.5 or higher) for both the group assignment (report and presentation) and the exam.

Improved versions of the group assignment or exam resits cannot receive grades higher than 6.0.

## 2.7 Deadlines and submitting assignments

Assignments must be submitted as PDF files. If multiple files are submitted, compress them into a .zip or .tar.gz archive. All assignments must be submitted via Canvas by the specified deadlines. Late submissions receive a grade of 1.

## 2.8 Plagiarism & fraud

The UvA Regulations Governing Fraud and Plagiarism apply fully to this course.

Students must ensure that all submitted work reflects their own knowledge and skills. Presenting work produced by others — including AI systems — as your own work constitutes academic misconduct.

Turnitin will be used to detect plagiarism.

### 2.8.1 Attributing code

Code may be inspired by lecture examples or online sources, but sources must be clearly cited.

Example:

```
# Code adapted from https://stackoverflow.com/...
```

You should only submit code that you understand and could explain to another student.

### 2.8.2 Use of AI tools

AI tools can support learning but must be used responsibly.

- AI tools may be used to clarify concepts, explain methods, or help debug code you have written yourself.
- Do not use AI tools to generate or substantially complete assignments that you submit.
- Do not blindly execute AI-generated code. You are responsible for understanding all submitted work.
- Prefer using the UvA AI Chat environment. Do not upload course datasets or sensitive information to commercial AI systems.
- If AI tools were used during your work, this must be disclosed.

Instructors may ask students to explain submitted code during tutorials or presentations.

## 2.9 Presence and participation

Students are expected to attend all meetings. Missing more than one tutorial may result in exclusion unless exceptional circumstances apply.

## 2.10 Language

All meetings and assignments will be conducted in English.

## 2.11 Teaching team

Anne Kroon and Saurabh Khanna teach the lectures. Roeland Dubèl teaches the tutorials.

Questions about lectures or assignments should be directed to Anne and Saurabh. Questions about tutorial logistics should be directed to Roeland.

Questions about coding errors should be addressed during tutorials rather than by email.

# Chapter 3

# Group assignment: Research report with code

This section describes the requirements of the group assignment.

### Teams

Form groups of around 3 to 4 students (4 is the maximum). Your tutorial teacher will help you with this.

### Dataset selection

Together with your group, you can select from one of the following datasets. The datasets are available on Canvas.

1. *News dataset*: This dataset contains internet news and user engagement metrics.
2. *Books dataset*: This dataset is a combination of two data sources, the Goodreads dataset and the 7k Books dataset.
3. *Recipe dataset*: This dataset contains data on numerous recipes.
4. *TV Series dataset*: This dataset contains details on TV series.
5. *Spotify Lyrics dataset*: This dataset contains Spotify tracks together with their song lyrics and metadata such as artist, genre, and popularity. The dataset is available on Kaggle.

When working with one of these datasets, it's crucial to note that not all columns need to be taken into consideration. You should choose the relevant variables yourself and provide justification in your assignment for including or excluding specific information.

The assignment will be centered around the choosen dataset. Your objective is to analyze the dataset in a data-scientific manner, and ultimately develop a recommender system.

The group assignment will be evaluated based on the following key components:

1. A written research report.
2. Demonstrated capability to write and apply code for preprocessing the data, conducting explorative data analyses, and building a simple knowledge-based or content-based recom-

mender system.

3. Quality and documentation of the provided code.

Please find detailed information about these components next.

## 3.1 Write a research report (50% of final grade)

Ensure that the research report is written in a clear and understandable manner. It is encouraged to include citations from relevant academic articles to support your analysis. Additionally, you may cite the book (Van Atteveldt, Trilling, & Calderon, 2022) to justify specific techniques or decisions made. The research report is structured into four sections:

### 3.1.1 Introduction

*Approximatly 0.5 - 1 page*

- Briefly explain why you chose the specific dataset.
- Highlight its relevance, potential insights, and why it is the most interesting for your group's objectives.
- State the overall aim of the Recommender System that you will develop in this assignment.
  - *For example, the Recommender System may focus on generating personalized book recommendations tailored to users' preferences, genres, and other users' behavior. Depending on the data you work with, this could be aimed at promoting reading habits or facilitating the sharing of relevant news within the community.*

### 3.1.2 Method section

*Approximatly 2.5 - 4.5 pages*

#### Data preprocessing

- Provide a detailed description of the final selection of variables, including their data types and any transformations applied.
- Explain the reasoning behind the approach taken for data preprocessing. Explain to your reader *why* these preprocessing steps are needed. You can think of the following reasons to apply preprocessing (see also the next section):
  - Visualizing the data.
  - Computing summary statistics.
- Explain clearly how the preprocessing steps are needed for the successful exectution of these techniques.

#### Exploratory Data Analysis (EDA)

- Describe the analytical strategy for EDA, including techniques such as visualization, summary statistics, or inductive methods like word clouds.

- Explain how these techniques will be used to gain insights into the dataset.
- Justify the selection of these techniques and their relevance to the dataset. For example:
  - Visualizing the data: This helps to identify outliers, trends, or patterns within the dataset.
  - Summary statistics: This helps in understanding the distribution, central tendencies, and variability of variables.
  - Inductive techniques like word clouds: Useful for textual data, so to uncover underlying themes and topics.

**Recommender System development**

- Specify the type of recommender system being developed and provide rationale for this choice.
- There are multiple decisions to consider at this stage, that you should elaborate on.
  - Specifically, explain whether you will build a knowledge-based or a content-based recommender system, and justify the choice of techniques used to create the recommendations. For example, you may choose to use cosine scores or soft-cosine scores.
  - Additionally, clarify which variables are included in the final recommendations and why they were selected.
  - Explain how the selected techniques will be used to build the recommender system.
- Justify the suitability of the chosen recommender system type for addressing the objectives of the assignment as described in the Introduction section.

### 3.1.3 Result section

*Approximatly 2.5 - 4.5 pages*

- Provide a comprehensive description of the dataset, including the number of observations and the types of variables present.
- Present the results of the Explorative Data Analysis (EDA) conducted on the dataset.
  - This may include insights gained from visualization, summary statistics, and/or inductive methods like word clouds. Describe patterns or trends that you have discovered during this analysis.
- Demonstrate the functionality of the developed recommender system.
  - Explain how the system works and provide examples of recommendations generated for various types of input.
  - This demonstration should showcase how the recommender system works in providing relevant recommendations based on user preferences or content characteristics.

### 3.1.4 Short conclusion

*Approximatly 0.5 - 1 page*

- Briefly summarize the key findings from the Exploratory Data Analysis (EDA).

- Reflect on the extent to which the original aim of the study specified in the Introduction was achieved: Was the recommendation system successful in meeting its objectives?
  - Please note that the aim is not necessarily to develop a perfect recommender system. You can provide critical insights here and suggest future strategies for enhancing its performance.

### 3.1.5 Quality of the writing, adherence to APA guidelines and references

Ensure the report is clearly written and adheres to APA guidelines. Including references to related academic work, if relevant, is appreciated.

### 3.1.6 Appendix: Division of labor

*Approximatly 0.5 page*

We believe it's essential for all team members to contribute equally to the group assignment. Therefore, in the Appendix, we request each team member to provide a brief description of their individual contributions to the group assignment. These contributions should be honestly reflected and cover the elements each team member worked on.

### 3.1.7 Appendix: AI use disclosure (if applicable)

If AI tools were used during the project, briefly describe how they were used. AI tools may be used to support learning (e.g., clarifying concepts or debugging code), but not to generate substantial parts of the submitted code or text.

If applicable, specify:

- which AI tool was used (e.g., UvA Chat AI),
- for which purpose it was used (e.g., explaining an error message or clarifying a concept),
- how the output was evaluated or modified by the group.

If no AI tools were used, please explicitly state: *"No AI tools were used in the preparation of this assignment."*

## 3.2 All code written in the project (35% of final grade)

### 3.2.1 Code for data preprocessing

- Code used to read in the data, explore the dataset, and inspect the relevant variables present.
- Your objective here is to select variables that might be of interest and usable later on.
- The code for cleaning and preprocessing your data is crucial here. Feature engineering plays a significant role in this step, considering the type of descriptive analysis you plan to conduct in the subsequent stages.

### 3.2.2 Code for Exploratory Data Analysis (EDA)

- Provide code to conduct exploratory data analysis (EDA) on the dataset.
- Ensure that the code yields a clear description of the data you will be analyzing, highlighting

key variables such as their data types, number of unique observations, mean values, and distribution characteristics.

- Utilize data visualization techniques, such as plotting, to visually represent relationships and patterns within the dataset.
- Apply various analytical techniques discussed in the first weeks of the course, such as word clouds, to uncover insights and themes present in the dataset.

### 3.2.3 Code to create a recommender system

- Provide the code to build a simple knowledge-based or content-based recommender system.
- Specifically, you have to provide code that builds a recommender system, based on the insights from week 3. It's up to you to decide whether you build a knowledge-based or content-based recommender system.
- Think about relevant features that you want to use in your algorithm design. Based on which features do you want to recommend content?

## 3.3 Quality of the code, and documentation (15% of final grade)

Make sure your code is well documented and understandable for people that see your code for the first time. This involves providing clear and concise comments within the code to explain its purpose and functionality. This requires providing clear and concise comments within the code to explain its purpose and functionality. Additionally, minimize code repetition and provide that is scalable if possible.

## 3.4 Handing in

One member of your group can hand in the group assignment until Monday, 6 May, 11:00 AM via Canvas. **Please compress all files into a single .zip or .tar.gz file**

Please include the following files:

## File formats

- A set of scripts used to preprocess and analyse the data, and to build the recommender system
- *You can hand in the answer to task 2* either *as one Jupyter Notebook-file, integrating code, output, and explanations* or *as one .py file containing the Python code and one PDF file with output and explanations.*
- The written research report should be .pdf format

*Note: You may, but do* ***not*** *have to create a shared (public) github repository where you store all code and documentation. In that case, please include the link to the github repository together with the written assignment to your tutorial teacher.*

***Good luck!!!***

# Chapter 4

# Course Schedule

This course is organised as follows. Lectures take place on Mondays and lab sessions on Tuesdays. During the lectures, key concepts are introduced from a conceptual and theoretical perspective, accompanied by examples of Python implementations.

Lab sessions begin with knowledge questions about the week's literature and lecture. Students then work on the assigned exercises and continue working on the group assignment once these are completed. The tutorial lecturer (Roeland) will circulate to help with questions. If several students encounter the same issue, it will be discussed plenarily. Active participation in the tutorial meetings is important to ensure that students understand the material and can work effectively on the assignments.

## Week 1: Course introduction & Working with textual data

### Monday, 30–3: Lecture (Anne)

In this first lecture, we introduce the course and its structure. We also explore how computational methods can be used in communication science by examining what we can learn from analysing textual data.

### Tuesday, 31–3: Lab session

✔ Read this manual and inspect the course Canvas page.
✔ Ensure that your computer is ready to use for the course.
✔ Read in advance: van Atteveldt and Peng (2018).
✔ Read in advance: Boumans and Trilling (2016).

During the first tutorial meeting we introduce the course goals and policies and apply the lecture material to a first text analysis exercise. We also revisit some basic concepts from CCS-1 and practise simple top-down and bottom-up approaches to text analysis.

Students will form groups of approximately four students for the group assignment. If you complete the in-class exercises early, you may begin exploring the dataset, inspecting relevant variables, and starting the data cleaning process.

## Week 2: Bottom-up approaches to text analysis

### Monday, 6–4: Easter Monday — no lecture

### Tuesday, 7–4: Lecture (Anne)

*This week the lecture takes place on Tuesday instead of Monday due to Easter Monday. There will be no separate lab session.*

✔ Read in advance: Chapter 10 "Text as data" Van Atteveldt et al. (2022).
✔ Read in advance: Kroon, Welbers, Trilling, and van Atteveldt (2024). Read from "From BoW to Large Language Models: Revolutions in text representations" until "The rise of the transformer" (pp. 147–149).

In this lecture we discuss different ways to represent text data, including Bag of Words, TF-IDF, and word embeddings.

Students will practise these concepts independently by transforming textual data into matrices using `sklearn`'s vectorizers, measuring similarity between documents, and experimenting with a simple LDA model.

## Week 3: Recommender systems

### Monday, 13–4: Lecture (Anne)

✔ Read in advance: Möller, Trilling, Helberger, and van Es (2018).
✔ Read in advance: Loecherbach and Trilling (2020).

This lecture introduces recommender systems. We discuss different types of recommendation approaches and how they rely on earlier concepts such as preprocessing text data, transforming texts into matrices, and calculating similarity between documents.

### Tuesday, 14–4: Lab session

During this lab session you will complete the first set of multiple-choice questions and practise building a simple recommender system.

## Week 4: Introduction to supervised machine learning Group presentation recommender system

### Monday, 20–4: Lecture (Saurabh)

✔ Read in advance: Vermeer, Araujo, Bernritter, and van Noort (2019).
✔ Read in advance: Meppelink et al. (2021).

This lecture introduces supervised machine learning and discusses its basic principles using practical examples and code.

### Tuesday, 21–4: Lab session

During this lab session groups will present their recommender system assignment.

### Deadline group project: Friday 24–4 at 23:59

The deadline for submitting the group assignment is **Friday 24–4 at 23:59**.

## Week 5: Teaching-free week

### Monday, 27–4: No lecture — UvA teaching-free week

### Tuesday, 28–4: No lab session — UvA teaching-free week

✔ Take a break

## Week 6

### Monday, 4–5: No lecture — Collectieve roostervrije dag

### Tuesday, 5–5: Liberation Day — no lab session

## Week 7: Validation in supervised machine learning

### Monday, 11–5: Lecture (Saurabh)

✔ Read in advance: Birkenmaier, Lechner, and Wagner (2023).

In this lecture we discuss validation techniques for supervised machine learning models.

### Tuesday, 12–5: Lab session

During the lab session we practise evaluating and validating machine learning models.

## Week 8: Coding in an academic context

### Monday, 18–5: Lecture (Saurabh)

✔ Read in advance: Hube, Fetahu, and Gadiraju (2019).
✔ Read in advance: Bender, Gebru, McMillan-Major, and Shmitchell (2021).

In this lecture we reflect on the responsible use of computational methods and machine learning in academic research.

### Tuesday, 19–5: Lab session

The final lab session focuses on responsible research coding and reproducible research practices.

## Week 9: Exam week

### Week of 1 June: Exam inspection

### Week of 8 June: Resit

### Week of 15 June: Resit inspection

# Literature


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Vermeer, S. A. M., Araujo, T., Bernritter, S. F., & van Noort, G. (2019). Seeing the wood for the trees: How machine learning can help firms in identifying relevant electronic word-of-mouth in social media. *International Journal of Research in Marketing*, *36*(3), 492–508. Retrieved 2024-02-29, from https://www.sciencedirect.com/science/article/pii/S0167811619300102 doi: 10.1016/j.ijresmar.2019.01.010